Data processing

Market Justice Preferences and the Dynamic Role of Objective and Subjective Status: A Longitudinal Multilevel Approach

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1 Presentation

This is the data preparation code for the paper "Market Justice Preferences and the Dynamic Role of Objective and Subjective Status: A Longitudinal Multilevel Approach". The prepared dataset is `ELSOC_Long_2016_2023_1.00.RData`.

2 Libraries

```
if (! require("pacman")) install.packages("pacman")

pacman::p_load(tidyverse,
               car,
               sjmisc,
               here,
               sjlabelled,
               SciViews,
               naniar,
               readxl,
               sjPlot,
               DIGCLASS)
```

```r
options(scipen=999)
rm(list = ls())
```

3 Data

```r
load(url("https://dataverse.harvard.edu/api/access/datafile/10797987"))

glimpse(elsoc_long_2016_2023)
```

4 Processing

```r
elsoc_long_2016_2023[elsoc_long_2016_2023 ==-999] <- NA
elsoc_long_2016_2023[elsoc_long_2016_2023 ==-888] <- NA
elsoc_long_2016_2023[elsoc_long_2016_2023 ==-777] <- NA
elsoc_long_2016_2023[elsoc_long_2016_2023 ==-666] <- NA

elsoc_long_2016_2023 <- elsoc_long_2016_2023 %>%
  mutate(just_pension = d02_01,
         just_educ = d02_02,
         just_salud = d02_03,
         merit_effort = c18_09,
         merit_talent = c18_10,
         age = m0_edad, m01,
         sex = m0_sexo,
         ideo = c15,
         protest = c08_02,
         retiro_2021 = m63_02,
         ess = d01_01,
         ess_og = d01_02,
         ess_hijos = d01_03) %>%
  as_tibble() %>%
  sjlabelled::drop_labels(., drop.na = FALSE)
```

```r
# Average index (1-5) of pension, education and healthcare market justice.
# Computed on the numeric items, before they are recoded to factors.
elsoc_long_2016_2023$mjp_index <- rowMeans(
  elsoc_long_2016_2023[, c("just_pension", "just_educ", "just_salud")],
  na.rm = FALSE
)

elsoc_long_2016_2023$mjp_index <- sjlabelled::set_label(elsoc_long_2016_2023$mjp_index
                         label = "Market justice preferences index")

sjmisc::descr(elsoc_long_2016_2023$mjp_index)
```

Basic descriptive statistics

```
 var    type                              label     n NA.prc mean   sd   se md
  dd numeric Market justice preferences index 17947  13.55    2 0.85 0.01  2
 trimmed   range iqr skew
     1.9 4 (1-5)   1 0.88
```

```r
elsoc_long_2016_2023$just_pension <- car::recode(elsoc_long_2016_2023$just_pension,
recodes = c("1='Strongly disagree'; 2='Disagree'; 3='Neither agree nor disagree'; 4='A
levels = c("Strongly disagree", "Disagree", "Neither agree nor disagree", "Agree", "St
as.factor = T)

elsoc_long_2016_2023$just_pension <- sjlabelled::set_label(elsoc_long_2016_2023$just_p
                         label = "Pension market justice")

elsoc_long_2016_2023$just_educ <- car::recode(elsoc_long_2016_2023$just_educ,
recodes = c("1='Strongly disagree'; 2='Disagree'; 3='Neither agree nor disagree'; 4='A
levels = c("Strongly disagree", "Disagree", "Neither agree nor disagree", "Agree", "St
as.factor = T)

elsoc_long_2016_2023$just_educ <- sjlabelled::set_label(elsoc_long_2016_2023$just_educ
                         label = "Education market justice")

elsoc_long_2016_2023$just_salud <- car::recode(elsoc_long_2016_2023$just_salud,
recodes = c("1='Strongly disagree'; 2='Disagree'; 3='Neither agree nor disagree'; 4='A
levels = c("Strongly disagree", "Disagree", "Neither agree nor disagree", "Agree", "St
```

```
as.factor = T)

elsoc_long_2016_2023$just_salud <- sjlabelled::set_label(elsoc_long_2016_2023$just_sal
                        label = "Healthcare market justice")
```

```
frq(elsoc_long_2016_2023$merit_effort)
```

```
Grado de acuerdo: Las personas son recompensadas por sus esfuerzos (x) <numeric>
# total N=20761 valid N=20629 mean=2.60 sd=1.02

Value |                          Label |    N | Raw % | Valid % | Cum. %
-------------------------------------------------------------------------
    1 |       Totalmente en desacuerdo | 2173 | 10.47 |   10.53 |  10.53
    2 |                   En desacuerdo | 9442 | 45.48 |   45.77 |  56.30
    3 | Ni en desacuerdo ni de acuerdo | 4037 | 19.45 |   19.57 |  75.87
    4 |                      De acuerdo | 4379 | 21.09 |   21.23 |  97.10
    5 |          Totalmente de acuerdo |  598 |  2.88 |    2.90 | 100.00
 <NA> |                          <NA> |  132 |  0.64 |    <NA> |   <NA>
```

```
frq(elsoc_long_2016_2023$merit_talent)
```

```
Grado de acuerdo: Las personas son recompensada por su inteligencia (x) <numeric>
# total N=20761 valid N=20631 mean=2.77 sd=1.03

Value |                          Label |    N | Raw % | Valid % | Cum. %
-------------------------------------------------------------------------
    1 |       Totalmente en desacuerdo | 1766 |  8.51 |    8.56 |   8.56
    2 |                   En desacuerdo | 8108 | 39.05 |   39.30 |  47.86
    3 | Ni en desacuerdo ni de acuerdo | 4571 | 22.02 |   22.16 |  70.02
    4 |                      De acuerdo | 5577 | 26.86 |   27.03 |  97.05
    5 |          Totalmente de acuerdo |  609 |  2.93 |    2.95 | 100.00
 <NA> |                          <NA> |  130 |  0.63 |    <NA> |   <NA>
```

```
elsoc_long_2016_2023 <- elsoc_long_2016_2023 %>%
  mutate(
    across(
      .cols = c(merit_effort, merit_talent),
      .fns = ~ car::recode(., recodes = c("1='Strongly disagree'; 2='Disagree';
```

```
                                          3='Neither agree nor disagree'; 4='Agree';
                                          5='Strongly agree'"),
                            levels = c("Strongly disagree", "Disagree", "Neither agree
                            as.factor = T)
    )
  )


elsoc_long_2016_2023$merit_effort <- sjlabelled::set_label(elsoc_long_2016_2023$merit_
                          label = "People are rewarded for their efforts")

elsoc_long_2016_2023$merit_talent <- sjlabelled::set_label(elsoc_long_2016_2023$merit_
                          label = "People are rewarded for their intelligence")
```

```
# sex
elsoc_long_2016_2023$sex <- car::recode(elsoc_long_2016_2023$sex,
                            recodes = c("1='Male'; 2='Female'"),
                            levels = c("Male", "Female"),
                            as.factor = T)

elsoc_long_2016_2023$sex <- sjlabelled::set_label(elsoc_long_2016_2023$sex,
                          label = "Gender")

# age
frq(elsoc_long_2016_2023$age)
```

```
Edad del entrevistado (x) <numeric>
# total N=20761 valid N=20761 mean=48.84 sd=15.43

Value |   N | Raw % | Valid % | Cum. %
--------------------------------------
   18 |  34 |  0.16 |    0.16 |   0.16
   19 |  72 |  0.35 |    0.35 |   0.51
   20 | 113 |  0.54 |    0.54 |   1.05
   21 | 156 |  0.75 |    0.75 |   1.81
   22 | 187 |  0.90 |    0.90 |   2.71
   23 | 231 |  1.11 |    1.11 |   3.82
   24 | 262 |  1.26 |    1.26 |   5.08
   25 | 324 |  1.56 |    1.56 |   6.64
```

26	318	1.53	1.53	8.17
27	312	1.50	1.50	9.68
28	361	1.74	1.74	11.42
29	345	1.66	1.66	13.08
30	365	1.76	1.76	14.84
31	364	1.75	1.75	16.59
32	369	1.78	1.78	18.37
33	377	1.82	1.82	20.18
34	384	1.85	1.85	22.03
35	373	1.80	1.80	23.83
36	444	2.14	2.14	25.97
37	373	1.80	1.80	27.76
38	411	1.98	1.98	29.74
39	365	1.76	1.76	31.50
40	386	1.86	1.86	33.36
41	391	1.88	1.88	35.24
42	395	1.90	1.90	37.15
43	379	1.83	1.83	38.97
44	361	1.74	1.74	40.71
45	370	1.78	1.78	42.49
46	410	1.97	1.97	44.47
47	411	1.98	1.98	46.45
48	405	1.95	1.95	48.40
49	403	1.94	1.94	50.34
50	451	2.17	2.17	52.51
51	417	2.01	2.01	54.52
52	462	2.23	2.23	56.75
53	413	1.99	1.99	58.74
54	463	2.23	2.23	60.97
55	495	2.38	2.38	63.35
56	498	2.40	2.40	65.75
57	449	2.16	2.16	67.91
58	468	2.25	2.25	70.17
59	461	2.22	2.22	72.39
60	481	2.32	2.32	74.70
61	397	1.91	1.91	76.61
62	393	1.89	1.89	78.51
63	386	1.86	1.86	80.37
64	355	1.71	1.71	82.08
65	362	1.74	1.74	83.82
66	316	1.52	1.52	85.34

67	335	1.61	1.61	86.96
68	253	1.22	1.22	88.17
69	260	1.25	1.25	89.43
70	282	1.36	1.36	90.79
71	268	1.29	1.29	92.08
72	230	1.11	1.11	93.18
73	235	1.13	1.13	94.32
74	241	1.16	1.16	95.48
75	245	1.18	1.18	96.66
76	180	0.87	0.87	97.52
77	152	0.73	0.73	98.26
78	105	0.51	0.51	98.76
79	98	0.47	0.47	99.23
80	62	0.30	0.30	99.53
81	40	0.19	0.19	99.73
82	19	0.09	0.09	99.82
83	9	0.04	0.04	99.86
84	11	0.05	0.05	99.91
85	4	0.02	0.02	99.93
86	2	0.01	0.01	99.94
87	2	0.01	0.01	99.95
88	3	0.01	0.01	99.97
89	3	0.01	0.01	99.98
90	2	0.01	0.01	99.99
91	1	0.00	0.00	100.00
92	1	0.00	0.00	100.00
<NA>	0	0.00	<NA>	<NA>

```
elsoc_long_2016_2023$aget <-
  factor(car::recode(elsoc_long_2016_2023$age,
                     "18:29=1;30:49=2;50:64=3;65:150=4"),
         labels = c('18-29', '30-49', '50-64', '65 or more'))
elsoc_long_2016_2023$aget <-
  sjlabelled::set_label(elsoc_long_2016_2023$aget,
                        label = c("Age groups"))


# political indentification

frq(elsoc_long_2016_2023$ideo)
```

```
Autoubicacion escala izquierda-derecha (x) <numeric>
# total N=20761 valid N=20443 mean=7.39 sd=3.96

Value |            Label |    N | Raw % | Valid % | Cum. %
--------------------------------------------------------------
    0 |      0 Izquierda | 1146 |  5.52 |    5.61 |   5.61
    1 |                1 |  386 |  1.86 |    1.89 |   7.49
    2 |                2 |  625 |  3.01 |    3.06 |  10.55
    3 |                3 | 1007 |  4.85 |    4.93 |  15.48
    4 |                4 | 1193 |  5.75 |    5.84 |  21.31
    5 |         5 Centro | 5360 | 25.82 |   26.22 |  47.53
    6 |                6 |  749 |  3.61 |    3.66 |  51.20
    7 |                7 |  707 |  3.41 |    3.46 |  54.65
    8 |                8 |  601 |  2.89 |    2.94 |  57.59
    9 |                9 |  177 |  0.85 |    0.87 |  58.46
   10 |       10 Derecha | 1135 |  5.47 |    5.55 |  64.01
   11 | 11 Independiente |  727 |  3.50 |    3.56 |  67.57
   12 |       12 Ninguno | 6630 | 31.93 |   32.43 | 100.00
 <NA> |             <NA> |  318 |  1.53 |    <NA> |   <NA>
```

```r
elsoc_long_2016_2023$ideo<-
factor(
  car::recode(
    elsoc_long_2016_2023$ideo,
    "c(11,12,-888,-999)='Does not identify';c(0,1,2,3,4)='Left';
     c(5)='Center';c(6,7,8,9,10)='Right'"
  ),
  levels = c('Left', 'Center', 'Right', 'Does not identify')
)

elsoc_long_2016_2023$ideo<- factor(elsoc_long_2016_2023$ideo,levels = levels(elsoc_lon

elsoc_long_2016_2023$ideo <-
sjlabelled::set_label(x = elsoc_long_2016_2023$ideo,
                      label = "Political identification")

frq(elsoc_long_2016_2023$ideo)
```

```
Political identification (x) <categorical>
# total N=20761 valid N=20443 mean=2.67 sd=1.17
```

```
Value             |    N | Raw % | Valid % | Cum. %
-------------------------------------------------------
Left              | 4357 | 20.99 |   21.31 |  21.31
Center            | 5360 | 25.82 |   26.22 |  47.53
Right             | 3369 | 16.23 |   16.48 |  64.01
Does not identify | 7357 | 35.44 |   35.99 | 100.00
<NA>              |  318 |  1.53 |    <NA> |   <NA>
```

```
# Subjective social status

frq(elsoc_long_2016_2023$ess)
```

```
Estatus Social Subjetivo: Donde se ubicaria ud. en la sociedad chilena (x) <numeric>
# total N=20761 valid N=20635 mean=4.44 sd=1.61

Value |                Label |    N | Raw % | Valid % | Cum. %
-----------------------------------------------------------------
    0 |  0 El nivel mas bajo |  451 |  2.17 |    2.19 |   2.19
    1 |                    1 |  529 |  2.55 |    2.56 |   4.75
    2 |                    2 | 1097 |  5.28 |    5.32 |  10.07
    3 |                    3 | 2644 | 12.74 |   12.81 |  22.88
    4 |                    4 | 4566 | 21.99 |   22.13 |  45.01
    5 |                    5 | 7933 | 38.21 |   38.44 |  83.45
    6 |                    6 | 1964 |  9.46 |    9.52 |  92.97
    7 |                    7 |  684 |  3.29 |    3.31 |  96.28
    8 |                    8 |  522 |  2.51 |    2.53 |  98.81
    9 |                    9 |   72 |  0.35 |    0.35 |  99.16
   10 | 10 El nivel mas alto |  173 |  0.83 |    0.84 | 100.00
 <NA> |                 <NA> |  126 |  0.61 |    <NA> |   <NA>
```

```
elsoc_long_2016_2023$ess <- as.numeric(elsoc_long_2016_2023$ess)

elsoc_long_2016_2023$ess <- sjlabelled::set_label(elsoc_long_2016_2023$ess,
                        label = "Subjective social status")

elsoc_long_2016_2023$ess_og <- as.numeric(elsoc_long_2016_2023$ess_og)

elsoc_long_2016_2023$ess_og <- sjlabelled::set_label(elsoc_long_2016_2023$ess_og,
```

```
                        label = "Family subjective social status")

elsoc_long_2016_2023$ess_hijos <- as.numeric(elsoc_long_2016_2023$ess_hijos)

elsoc_long_2016_2023$ess_hijos <- sjlabelled::set_label(elsoc_long_2016_2023$ess_hijos
                        label = "Son's subjective social status")
```

```
elsoc_long_2016_2023$educ <-
  car::recode(elsoc_long_2016_2023$m01,
              "c(1,2,3,4,5,6,7)=1;c(8,9,10)=2; c(-888,-999)=NA")
elsoc_long_2016_2023$educ <-
  factor(elsoc_long_2016_2023$educ,
         labels = c("Less than Universitary","Universitary"))

#reverse education, reference level is the highest level
#elsoc_long_2016_2023$educ <- forcats::fct_rev(elsoc_long_2016_2023$educ)

elsoc_long_2016_2023$educ <-
sjlabelled::set_label(x = elsoc_long_2016_2023$educ,
                      label = "Education")
sjmisc::frq(elsoc_long_2016_2023$educ)
```

```
Education (x) <categorical>
# total N=20761 valid N=20746 mean=1.19 sd=0.39

Value                   |     N | Raw % | Valid % | Cum. %
--------------------------------------------------------
Less than Universitary | 16786 | 80.85 |   80.91 |  80.91
Universitary           |  3960 | 19.07 |   19.09 | 100.00
<NA>                   |    15 |  0.07 |    <NA> |   <NA>
```

```
# Education in 4 categories
elsoc_long_2016_2023$educ_4 <-
  car::recode(elsoc_long_2016_2023$m01,
              "c(1,2,3)=1;c(4,5)=2;c(6,7)=3;c(8,9,10)=4;c(-888,-999)=NA")
elsoc_long_2016_2023$educ_4 <-
  factor(elsoc_long_2016_2023$educ_4,
         labels = c("Primary or less",
                    "Secondary",
```

```
                    "Technical",
                    "University or postgraduate"))

elsoc_long_2016_2023$educ_4 <-
sjlabelled::set_label(x = elsoc_long_2016_2023$educ_4,
                      label = "Education (4 categories)")
sjmisc::frq(elsoc_long_2016_2023$educ_4)
```

```
Education (4 categories) (x) <categorical>
# total N=20761 valid N=20746 mean=2.32 sd=1.02

Value                       |    N | Raw % | Valid % | Cum. %
----------------------------------------------------------------
Primary or less             | 4604 | 22.18 |   22.19 |  22.19
Secondary                   | 8814 | 42.45 |   42.49 |  64.68
Technical                   | 3368 | 16.22 |   16.23 |  80.91
University or postgraduate | 3960 | 19.07 |   19.09 | 100.00
<NA>                        |   15 |  0.07 |    <NA> |   <NA>
```

```
#Recoding of education to years based on casen 2017.
elsoc_long_2016_2023$educyear<- as.numeric(
  car::recode(elsoc_long_2016_2023$m01,
              "1=0;2=4.3;3=7.5;4=9.8;5=12.02;6=13.9;
               7=14.8;8=14.9;9=16.9;10=19.07;c(-888,-999)=NA",
              as.numeric = T))

elsoc_long_2016_2023$educyear <-
sjlabelled::set_label(x = elsoc_long_2016_2023$educyear,
                      label = "Education in years")

class(elsoc_long_2016_2023$educyear)
```

```
[1] "numeric"
```

```
sjmisc::frq(elsoc_long_2016_2023$educyear)
```

```
Education in years (x) <numeric>
# total N=20761 valid N=20746 mean=11.56 sd=3.99
```

```
Value |    N | Raw % | Valid % | Cum. %
--------------------------------------
 0.00 |  200 |  0.96 |    0.96 |   0.96
 4.30 | 2412 | 11.62 |   11.63 |  12.59
 7.50 | 1992 |  9.59 |    9.60 |  22.19
 9.80 | 2692 | 12.97 |   12.98 |  35.17
12.02 | 6122 | 29.49 |   29.51 |  64.68
13.90 |  749 |  3.61 |    3.61 |  68.29
14.80 | 2619 | 12.61 |   12.62 |  80.91
14.90 | 1197 |  5.77 |    5.77 |  86.68
16.90 | 2413 | 11.62 |   11.63 |  98.31
19.07 |  350 |  1.69 |    1.69 | 100.00
 <NA> |   15 |  0.07 |    <NA> |   <NA>
```

```r
elsoc_long_2016_2023 <- elsoc_long_2016_2023 %>%
 mutate(ola = case_when(ola == 1 ~ 2016,
                        ola == 2 ~ 2017,
                        ola == 3 ~ 2018,
                        ola == 4 ~ 2019,
                        ola == 5 ~ 2021,
                        ola == 6 ~ 2022,
                        ola == 7 ~ 2023))

# Impute midpoint of income ranges
elsoc_long_2016_2023$m30_rec <-
  as.numeric(car::recode(
    elsoc_long_2016_2023$m30,
    "1=110000;2=251000;3=305000;4=355000;5=400000;
     6=445000;7=490000;8=535000;9=585000;10=640000;11=700000;12=765000;
     13=845000;14=935000;15=1040000;16=1180000;17=1375000;18=1670000;
     19=2275000;20=2700000;NA=NA;c(-888,-999)=NA"
  ))

# Impute midpoint of income ranges (2021)
elsoc_long_2016_2023$m30b_rec <-
  as.numeric(car::recode(
    elsoc_long_2016_2023$m30b,
    "1=125000;2=300000;3=400000;4=575000;5=700000;NA=NA;c(-888,-999)=NA"
  ))
```

```
sjmisc::frq(elsoc_long_2016_2023$m30_rec)
```

```
x <numeric>
# total N=20761 valid N=4180 mean=784316.75 sd=633419.24

  Value |     N | Raw % | Valid % | Cum. %
----------------------------------------
 110000 |   332 |  1.60 |    7.94 |   7.94
 251000 |   259 |  1.25 |    6.20 |  14.14
 305000 |   267 |  1.29 |    6.39 |  20.53
 355000 |   219 |  1.05 |    5.24 |  25.77
 400000 |   296 |  1.43 |    7.08 |  32.85
 445000 |   225 |  1.08 |    5.38 |  38.23
 490000 |   326 |  1.57 |    7.80 |  46.03
 535000 |   145 |  0.70 |    3.47 |  49.50
 585000 |   253 |  1.22 |    6.05 |  55.55
 640000 |   125 |  0.60 |    2.99 |  58.54
 700000 |   184 |  0.89 |    4.40 |  62.94
 765000 |   196 |  0.94 |    4.69 |  67.63
 845000 |   129 |  0.62 |    3.09 |  70.72
 935000 |   118 |  0.57 |    2.82 |  73.54
1040000 |   249 |  1.20 |    5.96 |  79.50
1180000 |   175 |  0.84 |    4.19 |  83.68
1375000 |   124 |  0.60 |    2.97 |  86.65
1670000 |   219 |  1.05 |    5.24 |  91.89
2275000 |   191 |  0.92 |    4.57 |  96.46
2700000 |   148 |  0.71 |    3.54 | 100.00
   <NA> | 16581 | 79.87 |    <NA> |   <NA>
```

```
sjmisc::frq(elsoc_long_2016_2023$m30b_rec)
```

```
x <numeric>
# total N=20761 valid N=187 mean=350802.14 sd=198987.76

 Value |     N | Raw % | Valid % | Cum. %
----------------------------------------
125000 |    58 |  0.28 |   31.02 |  31.02
300000 |    45 |  0.22 |   24.06 |  55.08
```

```
400000 |    39 |  0.19 |   20.86 |  75.94
575000 |    18 |  0.09 |    9.63 |  85.56
700000 |    27 |  0.13 |   14.44 | 100.00
  <NA> | 20574 | 99.10 |    <NA> |   <NA>
```

```r
# Recode DK/DA of income to NA
elsoc_long_2016_2023$m29_rec <-
  as.numeric(car::recode(elsoc_long_2016_2023$m29, "c(-888,-999)=NA"))

# Replace missing continuous income with imputed midpoints
elsoc_long_2016_2023$m29_imp <- ifelse(!is.na(elsoc_long_2016_2023$m29_rec), elsoc_lon
summary(elsoc_long_2016_2023$m29_imp)
```

```
      Min.    1st Qu.     Median       Mean    3rd Qu.       Max.       NA's
         0     300000     500000     960976     800000 4000000000        882
```

```r
elsoc_long_2016_2023$m29_imp <- ifelse(is.na(elsoc_long_2016_2023$m29_imp), elsoc_long
summary(elsoc_long_2016_2023$m29_imp)
```

```
      Min.    1st Qu.     Median       Mean    3rd Qu.       Max.       NA's
         0     300000     500000     955290     800000 4000000000        695
```

```r
# Deflate to constant prices ----------------------------------------------
library(rvest)
url <- "https://si3.bcentral.cl/Siete/ES/Siete/Cuadro/CAP_PRECIOS/MN_CAP_PRECIOS/IPC_E

ipc <- url %>%
  read_html() %>%
  html_node("table") %>%
  html_table() %>%
  rename_with(., ~ tolower(gsub(".", "_", .x, fixed = TRUE))) %>%
  filter(serie == "Índice IPC General") %>%
  mutate(
    across(
      .cols = c(everything(), -serie),
      .fns = ~ as.numeric(str_replace(., ",", "."))
    )
  ) %>%
```

```
  select(-sel_) %>%
  pivot_longer(
    cols = -serie,
    names_to = "ano_mes",
    values_to = "ipc"
  ) %>%
  tidyr::separate(col = "ano_mes", into = c("mes", "ano"))

ipc <- ipc %>%
  filter(mes == "dic") %>%
  select(ano, ipc) %>%
  slice_head(n = 8) %>%
  filter(ano != "2020")

elsoc_long_2016_2023$ola <- as.factor(elsoc_long_2016_2023$ola)

elsoc_long_2016_2023 <- left_join(elsoc_long_2016_2023, ipc, by = c("ola" = "ano"))

frq(elsoc_long_2016_2023$ipc)
```

```
x <numeric>
# total N=20761 valid N=20761 mean=82.97 sd=10.48

 Value |    N | Raw % | Valid % | Cum. %
---------------------------------------
 72.29 | 2927 | 14.10 |   14.10 |  14.10
 73.93 | 2473 | 11.91 |   11.91 |  26.01
 75.83 | 3748 | 18.05 |   18.05 |  44.06
 78.10 | 3417 | 16.46 |   16.46 |  60.52
 86.20 | 2740 | 13.20 |   13.20 |  73.72
 97.21 | 2730 | 13.15 |   13.15 |  86.87
101.04 | 2726 | 13.13 |   13.13 | 100.00
  <NA> |    0 |  0.00 |    <NA> |   <NA>
```

```
# Build household size ----------------------------------

elsoc_long_2016_2023 <- elsoc_long_2016_2023 %>%
  mutate(
    n_hogar = dplyr::case_when(
```

```
      ola == "2016" ~ nhogar1,
      ola == "2017" ~ m46_nhogar,
      ola == "2018" ~ m54,
      ola == "2019" ~ m54,
      ola == "2021" ~ m54,
      ola == "2022" ~ m54,
      ola == "2023" ~ m54,
      TRUE ~ NA_real_
    )
  )

# Household size (m54) was not asked in 2022: carry it over from the same
# respondent's adjacent waves (2021, otherwise 2023). Without this, equivalised
# income and its deciles are NA for the whole 2022 wave and na.omit() drops it.
elsoc_long_2016_2023 <- elsoc_long_2016_2023 %>%
  group_by(idencuesta) %>%
  mutate(
    n_hogar = if_else(
      ola == "2022" & is.na(n_hogar),
      coalesce(n_hogar[ola == "2021"][1], n_hogar[ola == "2023"][1]),
      n_hogar
    )
  ) %>%
  ungroup()

sjmisc::frq(elsoc_long_2016_2023$n_hogar)
```

```
x <numeric>
# total N=20761 valid N=20278 mean=3.14 sd=1.58

Value |    N | Raw % | Valid % | Cum. %
--------------------------------------
    1 | 2894 | 13.94 |   14.27 |  14.27
    2 | 5017 | 24.17 |   24.74 |  39.01
    3 | 4737 | 22.82 |   23.36 |  62.37
    4 | 4129 | 19.89 |   20.36 |  82.73
    5 | 2068 |  9.96 |   10.20 |  92.93
    6 |  836 |  4.03 |    4.12 |  97.06
    7 |  337 |  1.62 |    1.66 |  98.72
    8 |  158 |  0.76 |    0.78 |  99.50
```

```
    9 |   48 |  0.23 |    0.24 |  99.73
   10 |   30 |  0.14 |    0.15 |  99.88
   11 |   10 |  0.05 |    0.05 |  99.93
   12 |    8 |  0.04 |    0.04 |  99.97
   13 |    2 |  0.01 |    0.01 |  99.98
   14 |    2 |  0.01 |    0.01 |  99.99
   15 |    2 |  0.01 |    0.01 | 100.00
 <NA> |  483 |  2.33 |    <NA> |   <NA>
```

```
# Recode DK/DA of household size to NA
elsoc_long_2016_2023$n_hogar_r <- car::recode(elsoc_long_2016_2023$n_hogar, "c(-888,-9
sjmisc::frq(elsoc_long_2016_2023$n_hogar_r)
```

```
x <numeric>
# total N=20761 valid N=20278 mean=3.14 sd=1.58

Value |    N | Raw % | Valid % | Cum. %
--------------------------------------
    1 | 2894 | 13.94 |   14.27 |  14.27
    2 | 5017 | 24.17 |   24.74 |  39.01
    3 | 4737 | 22.82 |   23.36 |  62.37
    4 | 4129 | 19.89 |   20.36 |  82.73
    5 | 2068 |  9.96 |   10.20 |  92.93
    6 |  836 |  4.03 |    4.12 |  97.06
    7 |  337 |  1.62 |    1.66 |  98.72
    8 |  158 |  0.76 |    0.78 |  99.50
    9 |   48 |  0.23 |    0.24 |  99.73
   10 |   30 |  0.14 |    0.15 |  99.88
   11 |   10 |  0.05 |    0.05 |  99.93
   12 |    8 |  0.04 |    0.04 |  99.97
   13 |    2 |  0.01 |    0.01 |  99.98
   14 |    2 |  0.01 |    0.01 |  99.99
   15 |    2 |  0.01 |    0.01 | 100.00
 <NA> |  483 |  2.33 |    <NA> |   <NA>
```

```
# Equivalised income (square-root scale) ------------------------------------

# Select a reference IPC (e.g., December 2023 as base)
ipc_ref <- elsoc_long_2016_2023 %>%
```

```
  filter(ola == "2023") %>%
  summarise(ipc_ref = median(ipc, na.rm = TRUE)) %>%
  pull(ipc_ref)

elsoc_long_2016_2023 <- elsoc_long_2016_2023 %>%
  mutate(
    # Deflate household income to constant prices (base = 2023 Dec)
    inc_hh_real = m29_imp * (ipc_ref / ipc),

    # Equivalence scale: square-root of household size
    eq_size_sqrt = if_else(!is.na(n_hogar_r) & n_hogar_r > 0,
                           sqrt(as.numeric(n_hogar_r)),
                           NA_real_),

    # Equivalised household income
    inc_eq_real_sqrt = inc_hh_real / eq_size_sqrt,

    # Log equivalised income (for modelling)
    ln_inc_eq_real_sqrt = if_else(!is.na(inc_eq_real_sqrt) & inc_eq_real_sqrt > 0,
                                  log(inc_eq_real_sqrt),
                                  NA_real_)
  )

sjmisc::descr(elsoc_long_2016_2023$m29_imp)
```

```
## Basic descriptive statistics

 var    type label     n NA.prc     mean       sd       se     md  trimmed
  dd numeric    dd 20066   3.35 955289.7 28330720 199998.7 500000 576741.8
                     range    iqr   skew
 4000000000 (0-4000000000) 500000 140.23
```

```
sjmisc::descr(elsoc_long_2016_2023$inc_eq_real_sqrt)
```

```
## Basic descriptive statistics

 var    type label     n NA.prc     mean       sd       se     md  trimmed
```

```
  dd numeric    dd 19611   5.54 761225.2 28308283 202145.3 369713 423402.6
                           range      iqr   skew
 3953303038.93 (0-3953303038.93) 379551.1 138.86
```

```
sjmisc::descr(elsoc_long_2016_2023$ln_inc_eq_real_sqrt)
```

```
## Basic descriptive statistics

 var    type label     n NA.prc  mean   sd   se    md trimmed             range
  dd numeric    dd 19515      6 12.87 0.78 0.01 12.83   12.84 12.92 (9.18-
22.1)
       iqr skew
 0.9644817 0.44
```

```
# Continuous income: equivalised real income in millions of CLP (base = 2023 Dec)
elsoc_long_2016_2023$inc_eq_mill <- elsoc_long_2016_2023$inc_eq_real_sqrt / 1000000

elsoc_long_2016_2023$inc_eq_mill <-
  sjlabelled::set_label(x = elsoc_long_2016_2023$inc_eq_mill,
                        label = "Equivalised household income (millions CLP, Dec 2023)
sjmisc::descr(elsoc_long_2016_2023$inc_eq_mill)
```

```
## Basic descriptive statistics

 var    type                                                  label     n NA.prc
  dd numeric Equivalised household income (millions CLP, Dec 2023) 19611   5.54
 mean    sd  se   md trimmed             range       iqr   skew
 0.76 28.31 0.2 0.37    0.42 3953.3 (0-3953.3) 0.3795511 138.86
```

```
# Original per-capita measure for comparison -------------
elsoc_long_2016_2023$ing_pc <- elsoc_long_2016_2023$m29_imp / elsoc_long_2016_2023$n_h
sjmisc::descr(elsoc_long_2016_2023$ing_pc)
```

```
## Basic descriptive statistics
```

```
 var    type label     n NA.prc     mean       sd       se     md  trimmed
  dd numeric    dd 19611   5.54 398714.1 14326811 102305.7 180000 215328.7
                     range      iqr  skew
 2000000000 (0-2000000000) 229666.7 138.7
```

```
# Income deciles (within-wave, based on equivalised income) -------------------

elsoc_long_2016_2023 <- elsoc_long_2016_2023 %>%
  group_by(ola) %>%
  mutate(decile_eq = ntile(inc_eq_real_sqrt, 10)) %>%
  ungroup()

frq(elsoc_long_2016_2023$decile_eq)
```

```
x <integer>
# total N=20761 valid N=19611 mean=5.50 sd=2.87

Value |    N | Raw % | Valid % | Cum. %
---------------------------------------
    1 | 1965 |  9.46 |   10.02 |  10.02
    2 | 1963 |  9.46 |   10.01 |  20.03
    3 | 1962 |  9.45 |   10.00 |  30.03
    4 | 1962 |  9.45 |   10.00 |  40.04
    5 | 1962 |  9.45 |   10.00 |  50.04
    6 | 1961 |  9.45 |   10.00 |  60.04
    7 | 1960 |  9.44 |    9.99 |  70.04
    8 | 1960 |  9.44 |    9.99 |  80.03
    9 | 1958 |  9.43 |    9.98 |  90.02
   10 | 1958 |  9.43 |    9.98 | 100.00
 <NA> | 1150 |  5.54 |    <NA> |   <NA>
```

```
# Include missing cases for income groups
elsoc_long_2016_2023$decile_eq1 <- if_else(is.na(elsoc_long_2016_2023$decile_eq), "DNA
elsoc_long_2016_2023$decile_eq1 <- factor(elsoc_long_2016_2023$decile_eq1, levels = c(

sjmisc::frq(elsoc_long_2016_2023$decile_eq1)
```

```
x <categorical>
# total N=20761 valid N=20761 mean=5.80 sd=3.06
```

```
Value |    N | Raw % | Valid % | Cum. %
--------------------------------------
1     | 1965 |  9.46 |    9.46 |   9.46
2     | 1963 |  9.46 |    9.46 |  18.92
3     | 1962 |  9.45 |    9.45 |  28.37
4     | 1962 |  9.45 |    9.45 |  37.82
5     | 1962 |  9.45 |    9.45 |  47.27
6     | 1961 |  9.45 |    9.45 |  56.72
7     | 1960 |  9.44 |    9.44 |  66.16
8     | 1960 |  9.44 |    9.44 |  75.60
9     | 1958 |  9.43 |    9.43 |  85.03
10    | 1958 |  9.43 |    9.43 |  94.46
DNA   | 1150 |  5.54 |    5.54 | 100.00
<NA>  |    0 |  0.00 |    <NA> |   <NA>
```

```
frq(elsoc_long_2016_2023$ola)  # ok
```

```
x <character>
# total N=20761 valid N=20761 mean=2019.34 sd=2.37

Value |    N | Raw % | Valid % | Cum. %
--------------------------------------
 2016 | 2927 | 14.10 |   14.10 |  14.10
 2017 | 2473 | 11.91 |   11.91 |  26.01
 2018 | 3748 | 18.05 |   18.05 |  44.06
 2019 | 3417 | 16.46 |   16.46 |  60.52
 2021 | 2740 | 13.20 |   13.20 |  73.72
 2022 | 2730 | 13.15 |   13.15 |  86.87
 2023 | 2726 | 13.13 |   13.13 | 100.00
 <NA> |    0 |  0.00 |    <NA> |   <NA>
```

```
elsoc_long_2016_2023 <- elsoc_long_2016_2023 %>%
  mutate(
    across(
      .cols = c(decile_eq, decile_eq1),
      .fns = ~ as.factor(.)
    )
  )
```

```r
# Income groups: Bottom (deciles 1-4), Middle (5-9), Top (10)
elsoc_long_2016_2023 <- elsoc_long_2016_2023 %>%
  mutate(
    inc_group = case_when(
      as.character(decile_eq) %in% as.character(1:4)  ~ "Bottom",
      as.character(decile_eq) %in% as.character(5:9)  ~ "Middle",
      as.character(decile_eq) == "10"                 ~ "Top"),
    inc_group = factor(inc_group, levels = c("Bottom", "Middle", "Top")),

    # Same grouping, keeping missing cases as "DNA"
    inc_group1 = case_when(
      as.character(decile_eq1) %in% as.character(1:4) ~ "Bottom",
      as.character(decile_eq1) %in% as.character(5:9) ~ "Middle",
      as.character(decile_eq1) == "10"                ~ "Top",
      as.character(decile_eq1) == "DNA"               ~ "DNA"),
    inc_group1 = factor(inc_group1, levels = c("Bottom", "Middle", "Top", "DNA"))
  )

elsoc_long_2016_2023$inc_group <-
  sjlabelled::set_label(x = elsoc_long_2016_2023$inc_group,
                        label = "Income group (deciles)")
elsoc_long_2016_2023$inc_group1 <-
  sjlabelled::set_label(x = elsoc_long_2016_2023$inc_group1,
                        label = "Income group (deciles), including missing")

sjmisc::frq(elsoc_long_2016_2023$inc_group)
```

```
Income group (deciles) (x) <categorical>
# total N=20761 valid N=19611 mean=1.70 sd=0.64

Value  |    N | Raw % | Valid % | Cum. %
---------------------------------------
Bottom | 7852 | 37.82 |   40.04 |  40.04
Middle | 9801 | 47.21 |   49.98 |  90.02
Top    | 1958 |  9.43 |    9.98 | 100.00
<NA>   | 1150 |  5.54 |    <NA> |   <NA>
```

```
sjmisc::frq(elsoc_long_2016_2023$inc_group1)
```

```
Income group (deciles), including missing (x) <categorical>
# total N=20761 valid N=20761 mean=1.83 sd=0.81

Value  |    N | Raw % | Valid % | Cum. %
---------------------------------------
Bottom | 7852 | 37.82 |   37.82 |  37.82
Middle | 9801 | 47.21 |   47.21 |  85.03
Top    | 1958 |  9.43 |    9.43 |  94.46
DNA    | 1150 |  5.54 |    5.54 | 100.00
<NA>   |    0 |  0.00 |    <NA> |   <NA>
```

```
# Reshape long to wide

df_study1_long <- elsoc_long_2016_2023 %>%
  select(idencuesta,
         ola,
         muestra,
         ponderador_long_total,
         segmento,
         estrato,
         just_pension,
         just_educ,
         just_salud,
         mjp_index,
         merit_effort,
         merit_talent,
         educ,
         educ_4,
         educyear,
         sex,
         age,
         aget,
         ideo,
         ess_og,
         ess_hijos,
         ess,
         decile_eq,
```

```
         decile_eq1,
         inc_group,
         inc_group1,
         inc_eq_real_sqrt,
         ln_inc_eq_real_sqrt,
         inc_eq_mill)

vars_invariantes <- c("age", "aget",
                      "sex",
                      "ideo")

df_study1_long_fix <- df_study1_long %>%
  arrange(idencuesta, muestra, ola) %>%
  group_by(idencuesta, muestra) %>%
  mutate(
    across(
      all_of(vars_invariantes),
      ~ {
        v1 <- .x[ola == 1 & !is.na(.x)]
        if (length(v1) > 0) v1[1] else .x
      }
    )
  ) %>%
  ungroup()

frq(df_study1_long_fix$ola) # ok
```

```
x <character>
# total N=20761 valid N=20761 mean=2019.34 sd=2.37

Value |    N | Raw % | Valid % | Cum. %
--------------------------------------
 2016 | 2927 | 14.10 |   14.10 |  14.10
 2017 | 2473 | 11.91 |   11.91 |  26.01
 2018 | 3748 | 18.05 |   18.05 |  44.06
 2019 | 3417 | 16.46 |   16.46 |  60.52
 2021 | 2740 | 13.20 |   13.20 |  73.72
 2022 | 2730 | 13.15 |   13.15 |  86.87
 2023 | 2726 | 13.13 |   13.13 | 100.00
 <NA> |    0 |  0.00 |    <NA> |   <NA>
```

```r
waves_use <- c("2016","2017","2018","2019","2022","2023")

df_study1_long_fix <- df_study1_long_fix %>%
  filter(ola %in% waves_use)
```

```r
df_unbalanced <- df_study1_long_fix

ids_t7 <- df_study1_long_fix %>%
  filter(ola == "2023") %>%
  distinct(idencuesta) %>%
  pull(idencuesta)

df_t7 <- df_study1_long_fix %>%
  filter(idencuesta %in% ids_t7)

df_ids_balanced <- df_t7 %>%
  group_by(idencuesta) %>%
  summarise(
    n_olas_distintas = n_distinct(ola),
    .groups = "drop"
  ) %>%
  filter(n_olas_distintas == length(waves_use)) %>%   # tiene las 6 olas
  pull(idencuesta)

df_balanced <- df_t7 %>%
  filter(idencuesta %in% df_ids_balanced)

# Número de individuos
n_distinct(df_unbalanced$idencuesta)
```

```
[1] 4447
```

```r
n_distinct(df_balanced$idencuesta)
```

```
[1] 1328
```

```r
# Distribución por ola
df_unbalanced %>% count(ola)
```

```
# A tibble: 6 x 2
  ola       n
  <chr> <int>
1 2016   2927
2 2017   2473
3 2018   3748
4 2019   3417
5 2022   2730
6 2023   2726
```

```r
df_balanced   %>% count(ola)
```

```
# A tibble: 6 x 2
  ola       n
  <chr> <int>
1 2016   1328
2 2017   1328
3 2018   1328
4 2019   1328
5 2022   1328
6 2023   1328
```

```r
df_study1 <- df_balanced %>%
  select(-muestra) %>%
  # na.omit() %>%

  mutate(ola = case_when(ola == "2016" ~ 1,
                         ola == "2017" ~ 2,
                         ola == "2018" ~ 3,
                         ola == "2019" ~ 4,
                         ola == "2022" ~ 5,
                         ola == "2023" ~ 6)) %>%
  mutate(ola = as.factor(ola),
         ola_num = as.numeric(ola),
         ola_2=as.numeric(ola)^2)
```

```r
df_study1 <- df_study1 %>%
  group_by(idencuesta) %>%              # Agrupar por el identificador del participante
  mutate(n_participaciones = n()) %>%  # Contar el número de filas (participaciones) p
  ungroup()

df_study1 <- df_study1 %>% filter(n_participaciones>1)
```

```r
df_study1$merit_effort <- as_numeric(df_study1$merit_effort)
df_study1$merit_talent <- as_numeric(df_study1$merit_talent)

df_study1$decile_eq <- as_numeric(df_study1$decile_eq)
df_study1$dummy_decile10 <- if_else(df_study1$decile_eq >= 10, 1, 0)
df_study1$dummy_educ <- if_else(df_study1$educ == "Universitary", 1, 0)

df_study1$age_2 <- (df_study1$age)^2

df_study1 <- df_study1 %>%
  mutate(ola = case_when(ola == 1 ~ "2016",
                         ola == 2 ~ "2017",
                         ola == 3 ~ "2018",
                         ola == 4 ~ "2019",
                         ola == 5 ~ "2022",
                         ola == 6 ~ "2023"),
         ola = factor(ola, levels = c("2016",
                                      "2017",
                                      "2018",
                                      "2019",
                                      "2022",
                                      "2023")))


df_study1 <- df_study1 %>% # WE and BE effects
  group_by(idencuesta) %>%
  mutate(merit_effort_mean = mean(merit_effort, na.rm = T),
         merit_effort_cwc = merit_effort - merit_effort_mean,
         merit_talent_mean = mean(merit_talent, na.rm = T),
         merit_talent_cwc = merit_talent - merit_talent_mean,
         ess_mean = mean(ess, na.rm = T),
         ess_cwc = ess - ess_mean,
```

```
          ess_og_mean = mean(ess_og, na.rm = T),
          ess_og_cwc = ess_og - ess_og_mean,
          ess_hijos_mean = mean(ess_hijos, na.rm = T),
          ess_hijos_cwc = ess_hijos - ess_hijos_mean,
         dummy_educ_mean = mean(dummy_educ, na.rm = T),
         dummy_educ_cwc = dummy_educ - dummy_educ_mean,
         dummy_decile10_mean = mean(dummy_decile10, na.rm = T),
         dummy_decile10_cwc = dummy_decile10 - dummy_decile10_mean,
         educyear_mean = mean(educyear, na.rm = T),
         educyear_cwc = educyear - educyear_mean,
         ln_inc_eq_real_sqrt_mean = mean(ln_inc_eq_real_sqrt, na.rm = T),
         ln_inc_eq_real_sqrt_cwc = ln_inc_eq_real_sqrt - ln_inc_eq_real_sqrt_mean,
         decile_eq_mean = mean(decile_eq, na.rm = T),
         decile_eq_cwc = decile_eq - decile_eq_mean,
          ) %>%
  ungroup()


df_study1 <- df_study1 %>%
  group_by(idencuesta) %>%
  mutate(dummy_decile10_mean = as.integer(any(dummy_decile10 == 1, na.rm = TRUE)),
         dummy_educ_mean = as.integer(any(dummy_educ == 1, na.rm = TRUE))) %>%
  ungroup()

df_study1$just_pension <- as_factor(df_study1$just_pension)
df_study1$just_educ <- as_factor(df_study1$just_educ)
df_study1$just_salud <- as_factor(df_study1$just_salud)
```

```
# Subjective social status index: mean of own, family-of-origin and children's
# status (all 0-10 scales)
sss_vars <- c("ess", "ess_og", "ess_hijos")

# Internal consistency and inter-item correlations
psych::alpha(df_study1[, sss_vars])$total[c("raw_alpha", "std.alpha")]
```

```
 raw_alpha std.alpha
   0.66487 0.6827389
```

```r
cor(df_study1[, sss_vars], use = "pairwise.complete.obs")
```

```
                ess    ess_og ess_hijos
ess       1.0000000 0.5187860 0.4374827
ess_og    0.5187860 1.0000000 0.2968295
ess_hijos 0.4374827 0.2968295 1.0000000
```

```r
df_study1 <- df_study1 %>%
  group_by(idencuesta) %>%
  mutate(mov_dir      = ess - ess_og,
         mov_dir_mean = mean(mov_dir),
         mov_dir_cwc  = mov_dir - mov_dir_mean,
         mov_og       = ess - ess_og_mean,
         mov_og_mean  = mean(mov_og),
         mov_og_cwc   = mov_og - mov_og_mean) %>%
  ungroup()
```

```r
df_study1 <- df_study1 %>%
  mutate(mov_exp = ess_hijos - ess) %>%
  group_by(idencuesta) %>%
  mutate(mov_exp_mean = mean(mov_exp),
         mov_exp_cwc  = mov_exp - mov_exp_mean) %>%
  ungroup()
```

```r
df_study1 <- df_study1 %>%
  mutate(mov_inter = ess_hijos - ess - ess_og) %>%
  group_by(idencuesta) %>%
  mutate(mov_inter_mean = mean(mov_inter),
         mov_inter_cwc  = mov_inter - mov_inter_mean) %>%
  ungroup()
```

```r
# Children's SSS minus family-of-origin SSS: positive values = higher
# expected social mobility
df_study1 <- df_study1 %>%
  mutate(mov_exp_og = ess_hijos - ess_og) %>%
  group_by(idencuesta) %>%
  mutate(mov_exp_og_mean = mean(mov_exp_og),
         mov_exp_og_cwc  = mov_exp_og - mov_exp_og_mean) %>%
  ungroup()
```

```r
# SSS trajectory within each person-wave: family of origin (t = 1), own (t = 2)
# and children (t = 3). The OLS slope of ess ~ t over these three points is the
# trajectory (positive = upward). NA if any of the three measures is missing
df_study1 <- df_study1 %>%
  rowwise() %>%
  mutate(ess_traj = {
    y <- c(ess_og, ess, ess_hijos)
    if (anyNA(y)) NA_real_ else unname(coef(lm(y ~ c(1, 2, 3)))[2])
  }) %>%
  ungroup()
```

5 Save and export

```r
save(df_study1_long,file = here::here("input/data/rob_income/df_study1_long_vf_rob_inc

save(df_study1_long_fix,file = here::here("input/data/rob_income/df_study1_long_fix_vf

save(df_unbalanced,file = here::here("input/data/rob_income/df_unbalanced_rob_income.R
save(df_balanced,file = here::here("input/data/rob_income/df_balanced_rob_income.RData

save(df_study1, file = here::here("input/data/proc/df_final.RData"))
```